

Small-Scale SOC Simulation – Project Plan

1. Project Overview

This project is a **Small-Scale Security Operations Center (SOC) Simulation** designed to replicate the environment and responsibilities of a **Junior Cybersecurity Analyst**. It will involve deploying a SIEM platform (Security Onion or Wazuh) within a **virtualized lab environment** using **UTM** on a MacBook, connecting multiple endpoint virtual machines via agents, and simulating various security events for detection, analysis, and incident response practice.

2. Problem Statement

Organizations face continuous cyber threats and require skilled analysts to detect, analyze, and respond to incidents in real time. However, new analysts often lack practical experience working with SIEM platforms, endpoint monitoring, and incident response workflows.

This project addresses that gap by **creating a realistic, self-contained SOC environment** where security events can be generated and analyzed, preparing for real-world junior analyst responsibilities.

3. Project Objectives

- Set up a **fully functional SIEM** (Security Onion or Wazuh) in a virtual lab.
- Deploy **multiple endpoint VMs** and configure endpoint agents.
- Collect and analyze logs from diverse sources (Windows, Linux, firewall).
- Simulate attacks and security incidents.
- Create **custom detection rules** for suspicious activity.
- Document incidents in a professional **Incident Response Playbook**.
- Demonstrate findings through dashboards, reports, and a final walkthrough.

4. Scope

In-Scope:

- SIEM deployment in UTM.
- Multiple endpoint systems (Windows + Linux).
- Open-source-only tools.
- Basic attack simulation and log ingestion.
- Documentation of environment, rules, and incidents.

Out-of-Scope:

- Integration with live enterprise networks.
- Handling real malware (only safe test files allowed).
- Cloud-based deployments.

5. Tools & Technologies

- **Virtualization:** UTM for macOS
 - **SIEM:** Security Onion or Wazuh
 - **Endpoint Agents:** Wazuh Agent, Beats (Winlogbeat/Filebeat), Syslog
 - **Operating Systems:** Windows 10/11 Eval, Ubuntu Linux Server/Desktop
 - **Attack Simulation:** Nmap, EICAR test file, basic brute-force scripts
 - **Documentation:** draw.io / Lucidchart, Markdown, PDF reports
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6. Timeline

Month 1 – Environment Setup

Goal: Have a functioning SIEM with connected endpoints and log flow.

- Install UTM on Mac and configure virtual networking.
 - Deploy Security Onion or Wazuh in a VM.
 - Create at least 2 endpoint VMs (Windows + Linux).
 - Install and configure endpoint agents.
 - Verify log ingestion.
 - Document setup and architecture.
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Month 2 – Detection & Simulation

Goal: Create alerts and simulate real-world attacks.

- Write 3–5 custom detection rules.
 - Simulate brute force, port scans, and malware download attempts.
 - Tune detection rules to reduce false positives.
 - Start initial incident response documentation.
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Month 3 – Incident Response & Final Deliverables

Goal: Produce a full incident response package and present findings.

- Create an Incident Response Playbook.
- Build SIEM dashboards for monitoring.

- Produce final incident reports with screenshots.
 - Record a walkthrough/demo video.
 - Submit final project document.
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7. Month 1 – Week-by-Week Plan

Week 1

- Install UTM on Mac.
- Download ISOs for SIEM and endpoint OS.
- Create base network diagram.

Week 2

- Deploy SIEM VM (Security Onion/Wazuh).
- Configure basic settings (IP, networking, storage).

Week 3

- Create endpoint VMs (Windows + Linux).
- Configure host-only or NAT networking in UTM.

Week 4

- Install endpoint agents.
 - Verify logs are visible in SIEM.
 - Test simple log events (failed logins, command history).
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8. Deliverables

1. **Technical Documentation** – Environment setup guide with screenshots.
 2. **Network Diagram** – Showing SIEM, endpoints, and log flow.
 3. **Detection Rules** – List of custom rules and their purpose.
 4. **Incident Response Playbook** – Steps for responding to detected threats.
 5. **Final Dashboard** – SIEM dashboard with active monitoring views.
 6. **Final Presentation/Demo Video** – Walkthrough of lab setup, detection, and response.
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9. Success Criteria

- All endpoint agents report to the SIEM.
- Custom rules trigger alerts for simulated attacks.
- Incident reports are clear, accurate, and professionally formatted.
- Final demo successfully shows the detection and response workflow.